B.E.SEM-VII

AIML in Healthcare (2023-2024)

Ch.6 Future of Healthcare using AI and ML

Q.1 Compare Evidence-Based Medicine (EBM) and Personalized Medicine.

	EBM	**Personalized Medicine**
Definition	EBM emphasizes the use of clinical evidence from well-designed research studies to guide medical decision-making. It aims to apply standardized treatment guidelines based on the best available scientific evidence for the average patient.	Personalized Medicine, also known as precision medicine, tailors medical care to individual characteristics, such as genetics, lifestyle, and environmental factors. It recognizes that not all patients are the same and that treatment should be personalized to the specific needs of each person.
Patient-Centered Approach	While EBM considers the best available evidence, it may not always take into account individual patient variations, preferences, or genetic factors.	Personalized Medicine is inherently patient-centered, focusing on individual differences and needs to deliver more effective and personalized treatments.
Data and Information	EBM relies on large-scale clinical trials and systematic reviews of existing research to establish standardized treatment guidelines.	Personalized Medicine incorporates a wide range of data, including genetic, genomic, proteomic, and

		environmental information, to tailor treatment decisions.
Target Population	EBM guidelines typically apply to broader patient populations and aim to provide standardized care that works for most individuals.	Personalized Medicine is particularly relevant for conditions with a genetic or molecular basis, where individual differences have a significant impact on treatment outcomes.
Treatment Selection	EBM typically relies on one-size-fits-all treatment guidelines based on the average response of a population to a particular therapy.	Personalized Medicine selects treatments based on an individual's genetic makeup, disease characteristics, and other personal factors to optimize therapeutic outcomes.
Examples	Guidelines for treating common conditions like hypertension, where standardized treatments are often effective for the majority of patients.	Targeted therapies for cancer, which are tailored to the specific genetic mutations present in a patient's tumor.
Challenges	EBM may not fully account for individual variations and can be limited by the quality and availability of evidence.	Personalized Medicine faces challenges related to data collection, interpretation, and access to targeted

		therapies, which may not be widely available or affordable.
Evolution	EBM continues to evolve with advancements in medical research, but it remains grounded in the principles of standardized care.	Personalized Medicine represents a more recent and rapidly evolving approach, driven by advances in genetics, genomics, and data science.

Evidence-Based Medicine relies on standardized guidelines developed from population-level research, while Personalized Medicine tailors healthcare to the unique characteristics and needs of individual patients, often incorporating genetic and molecular information.

Q.2 Define the fundamentals of Block chain Technology. Give various Applications of Block chain in Healthcare.

The blockchain is general terms is defined in following manner:

- A technology that permits transactions to be recorded permanently.
- A technology that cryptographically secure the system and chains data in chronological order.
- A technology that remove intermediaries and create trust through the algorithm.

Blockchain technology has several potential applications in healthcare, Some key applications include:

1. Medical Records Management:

- Blockchain can securely store and manage patients' electronic health records (EHRs). Patients have control over who can access their data, enhancing privacy and interoperability.

2. **Health Data Interoperability**:

- Blockchain can improve data sharing and interoperability among different healthcare providers and systems, ensuring accurate and real-time patient information.

3. **Clinical Trials**:

- Blockchain can streamline the management of clinical trial data, ensuring the integrity of data and making the process more transparent. It can also enhance patient consent management.

4. **Drug Traceability and Supply Chain**:

- Blockchain can be used to track pharmaceuticals through the supply chain, reducing the risk of counterfeit drugs and ensuring the authenticity and safety of medications.

5. **Telemedicine and Remote Monitoring**:

- Securely store and share telemedicine consultations and remote monitoring data. Patients and healthcare providers can access data from anywhere.

6. **Health Insurance and Claims Processing**:

- Streamline health insurance and claims processing by automating verification and payments, reducing fraud, and improving transparency.

7. **Identity Management**:

- Manage patient identities securely, reducing the risk of identity theft and ensuring accurate patient information.

8. **Drug Adherence and Prescription Management**:

- Encourage patient adherence to medication schedules by providing real-time tracking and monitoring. Blockchain can also verify prescription authenticity.

9. **Health Research and Data Sharing**:

- Researchers can securely access aggregated and anonymized health data for studies while ensuring patient privacy.

10. **Public Health and Disease Surveillance**:

- Enable real-time tracking and reporting of disease outbreaks, helping public health authorities respond more effectively.

11. **Genomic Data Storage and Sharing**:

- Securely store and share genomic data for research and personalized medicine applications.

12. **Donor Organ Tracking**:

- Improve the tracking of donated organs for transplantation, ensuring they are properly matched and handled.

13. **Pharmacy Benefit Management**:

- Optimize pharmacy benefit management by streamlining processes and ensuring accurate prescription information.

14. **Patient Consent Management**:

- Use blockchain for managing patient consent, allowing patients to control who accesses their data and for what purposes.

Blockchain in healthcare has the potential to enhance security, data integrity, and interoperability while empowering patients with more control over their health data. However, its widespread adoption still faces regulatory and technical challenges.